

# Baseball Realtime — Application Overview

## What the Application Is

Baseball Realtime is a live and historical MLB game viewer focused on presenting baseball as an event stream rather than a traditional static box score.

The application combines:

- Live MLB data ingestion
- Real-time event streaming
- Rich pitch-by-pitch visualization
- Timeline navigation
- Box score and player statistics
- Historical game replay
- Responsive game-centric UI

The overall design goal is to make a baseball game feel "alive" while still supporting deep inspection of plays, players, scoring events, and momentum shifts.

The application is designed for both:

- Live game monitoring
  - Post-game replay and analysis
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## High-Level Architecture

### Backend

#### Technology

- NestJS
- TypeScript
- MySQL
- WebSocket realtime streaming
- Polling-based MLB feed ingestion

#### Responsibilities

The backend:

- Polls MLB APIs for live game data
- Normalizes MLB feed structures

- Tracks pitch/event progression
- Builds incremental game frames
- Streams updates to connected clients
- Generates derived game state
- Produces box score DTOs
- Tracks inning/count/base state transitions
- Maintains branding/team metadata caches

## Realtime Pipeline

The poller:

1. Retrieves MLB live feed data
  2. Detects newly available plays/pitches
  3. Builds normalized realtime events
  4. Publishes events through WebSockets
  5. Allows clients to replay or continue from current state
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## Frontend

### Technology

- React
- TypeScript
- Vite
- CSS modules / custom CSS
- OpenAPI-generated API client

### Primary UX Goals

- Minimal latency
  - High information density
  - Fast visual scanning
  - Smooth replay experience
  - Timeline-driven navigation
  - Mobile-friendly responsive layout
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## Current Major Features

### 1. Daily Games View

The Daily Games page acts as the landing page for the application.

## Features

- Displays all MLB games for a selected date
- Previous/next date navigation
- Team logos and branding
- Live / scheduled / final game states
- Scoreboard presentation
- Selected game highlighting
- Persisted selected date

## Game Metadata

Each game card can display:

- Home/away teams
  - Current score
  - Inning state
  - Game status
  - Venue information
  - Start time
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## 2. Live Game Viewer

The Game Page is the primary experience.

It presents:

- Scoreboard
- Box score
- Timeline
- Pitch-by-pitch feed
- Realtime updates

all synchronized around the current game state.

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## 3. Realtime Pitch-by-Pitch Feed

The pitch feed is one of the core application features.

### Current Feed Model

The feed is ordered top-to-bottom chronologically.

The feed groups events into:

- Inning sections
- At-bat sections
- Individual pitch/play events
- Terminal events

## **Feed Event Types**

### **Inning Markers**

Clearly separated inning transitions.

Examples:

- Top 3rd
- Bottom 7th

### **Batter Headers**

At-bat grouping headers showing:

- Batter
- Pitcher
- Situation context

### **Pitch Events**

Individual pitch results:

- Ball
- Strike
- Foul
- In-play result
- Velocity/location information (when available)

### **Terminal Events**

At-bat-ending events visually emphasized.

Examples:

- Strikeout
- Walk
- Single
- Double
- Home run

- Ground out
- Sacrifice fly

## Feed Styling Improvements Already Implemented

- Improved inning separation
  - Reduced excess vertical spacing
  - Enhanced terminal event styling
  - Better distinction between scoring and inning markers
  - Chronological readability improvements
  - Consistent top-down event flow
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## 4. Historical Replay Mode

Completed games can be replayed as though they are live.

### Replay Features

- Sequential event playback
- Configurable replay speed
- Automatic scrolling behavior
- Live-feed simulation
- Preservation of inning/game state transitions

### Replay Speed Settings

Replay timing is configurable and persisted locally.

Supported delay presets include:

- 50 ms
  - 100 ms
  - 250 ms
  - 500 ms
  - 1000 ms
  - 1500 ms
  - 2000 ms
  - 3000 ms
  - 4000 ms
  - 5000 ms
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## 5. Game Timeline Navigation

A timeline component allows direct navigation to key game moments.

### Timeline Marker Types

#### Inning Markers

Jump directly to inning boundaries.

#### Scoring Markers

Jump directly to scoring plays.

### Current Timeline Improvements

- Accurate inning anchor scrolling
  - Batter-based scoring anchors
  - Correct scroll ancestor handling
  - Improved scroll alignment
  - Stable anchor ID generation
  - Timeline synchronization with feed state
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## 6. Box Score System

The application includes a detailed live box score implementation.

### Team Sections

Separate:

- Batting
- Bench
- Pitching

for each team.

### Batting Features

- Batting order sorting
- Starter grouping
- Bench separation
- Player substitutions nested beneath replaced starters

- Jersey numbers
- Position display
- Live stat updates

## Pitching Features

- Starting pitcher tracking
- Bullpen tracking
- Live pitching statistics

## Statistical Highlighting

Recently changed statistics can be visually highlighted.

Examples:

- RBI changes
  - Hit totals
  - Strikeout totals
  - Home runs
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# 7. Player Detail Views

The application supports player-centric presentation.

## Hitter Displays

Compact stat presentation:

- AVG
- OBP
- SLG

## Pitcher Displays

Compact stat presentation:

- IP
- ERA
- WHIP
- K

## Visual Features

- MLB headshots
  - Team branding
  - Compact card presentation
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## 8. Branding and Visual Identity

The UI heavily incorporates MLB branding.

### Current Branding Features

- Team logos
  - Team color accents
  - Watermark score displays
  - Dynamic scorecard presentation
  - Consistent panel styling
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## 9. Responsive Layout System

The layout system dynamically balances multiple synchronized panels.

### Layout Goals

- Keep scoreboard visible
- Prevent unnecessary page scrolling
- Allow feed scrolling within constrained panels
- Maintain synchronized panel heights

### Implemented Behaviors

- Dynamic height calculations
  - ResizeObserver-based layout updates
  - Independent feed scrolling containers
  - Stable left/right panel alignment
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## 10. Settings System

The application includes user-configurable settings.



## Current Settings

### Replay Delay

Persisted replay speed configuration.

### Stored Preferences

Examples:

- Replay timing
- Selected date

via localStorage.

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## 11. Realtime Synchronization

The application supports continuous realtime updates.

### Live Update Data Includes

- Inning
  - Half inning
  - Count
  - Base runners
  - Batter
  - Pitcher
  - Score
  - Hits/errors
  - Description text
  - Branding metadata
  - Game status
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## 12. API / DTO Layer

The backend exposes strongly typed DTOs.

### DTO Areas

- Game summaries
- Live updates
- Box score data

- Player statistics
- Timeline events
- Feed entries

The frontend consumes these through an OpenAPI-generated TypeScript client.

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## Current UX Direction

The current UX direction is moving toward:

- Richer at-bat visualization
- Reduced scrolling requirements
- State-oriented baseball storytelling
- More immersive replay experiences
- Event-centric navigation

A major active design concept is the evolution of the feed toward an "At Bat Card" model where each at-bat becomes a stateful object that transitions pitch-by-pitch.

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## Major Planned / In-Progress Features

### At-Bat Card System

A more advanced visualization model where:

- Each at-bat is represented as a persistent card
- The card transitions through pitch states
- Previous states remain inspectable
- Pitch progression becomes visual rather than purely textual

Potential tracked state includes:

- Inning
  - Score
  - Batter
  - Pitcher
  - Count
  - Runners on base
  - Outs
  - Pitch sequence
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## Advanced Timeline Features

Planned enhancements include:

- Timeline scrubber
  - Replay time-travel
  - Feed collapsing
  - Multi-game synchronization
  - Late-game focus mode
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## Expanded Statistics System

Planned additions include:

- League leaders pages
  - Advanced pitching metrics
  - Pitch mix analytics
  - Leverage analysis
  - Expanded player drill-downs
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## Shared / Social Features

Future concepts include:

- Shared game rooms
  - Collaborative viewing
  - Watch rules and alerts
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## Overall Product Direction

The application is evolving toward a hybrid of:

- Traditional scoreboard
- Live event terminal
- Digital scorebook
- Replay engine
- Baseball analytics viewer

with a strong emphasis on:

- immersion
- realtime responsiveness

- historical playback
- narrative presentation of baseball events
- efficient visual scanning for dedicated baseball fans.